

CICLOMÉTRICAS



Cycling Performance Analysis

Free and open-source desktop application
for cycling power data analysis

Compatible with .FIT and .TCX files · Strava Integration

Windows · Linux

GPL v3.0 License

v3.0 — June 2026

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1. What is Ciclométricas?

Ciclométricas is a free, open-source, cross-platform desktop application designed by and for cyclists who train with a power meter. It provides a deep, scientifically grounded analysis of power, heart rate, cadence, and pedal balance data, with tools that go far beyond what mainstream platforms offer.

Inspired by the methodology of leading tools such as Golden Cheetah, WKO5, TrainingPeaks, and Intervals.icu, Ciclométricas brings together in a single local application the Critical Power (CP) model, the Durability Index (DRI), fatigue resistance, W' Balance, training monotony, race readiness, a comprehensive cardiovascular health module, and more advanced analyses.

Project philosophy:

- **100% local:** your data is yours. Everything is processed locally.
- **Scientifically rigorous:** based on peer-reviewed publications (Monod-Scherrer, Skiba, Foster, Coggan, Friel).
- **Free and open:** no subscriptions, no limits, no ads.
- **Cross-platform:** compatible with .FIT (Garmin, Wahoo, Hammerhead...) and .TCX. Direct sync with Strava.
- **Open Source:** licensed under GPL v3.0. Source code available on GitHub.

2. What's New in Version 2.0

Version 2.0 introduces significant improvements in analysis, stability, and new health tracking tools:

► Health Module (new)

New comprehensive cardiovascular and body composition metrics module. Track resting HR, HRV, blood pressure, weight, body fat, subcutaneous fat, and readiness score. Includes three interactive charts (cardiovascular, candlestick-style blood pressure, body composition), a historical table with inline editing/deletion, and readiness normalization from multiple sources (Garmin, Whoop, Oura, COROS, Elite HRV, manual).

► HR Zones Based on LTHR (Friel)

Heart rate zones are now calculated from LTHR (Lactate Threshold Heart Rate) instead of maximum HR, following Joe Friel's updated model with 7 zones (Z1 → Z5c).

► Left/Right Pedal Balance

New left/right power balance analysis, available in the detail view of each activity. Visual bar with symmetry classification (balanced, slight asymmetry, notable asymmetry) and dominant leg identification. Compatible with dual-sensor power meters.

► Efficiency Trends (3 Independent Axes)

The efficiency trends chart now displays three metrics with independent Y axes: EF (NP/HR) in emerald, VF (NP/Pavg) in amber, and Pw:Hr (power/HR decoupling) in cyan. Each metric has its own scale to avoid visual confusion from shared scales.

► Strava Import Stability

Complete rewrite of the Strava import system. The worker now uses native Python threading with QTimer polling, eliminating freezing issues that could occur when navigating after a bulk import.

► **Improved TTE for Sub-CP Power**

The Time to Exhaustion (TTE) calculator now provides detailed zone-based estimates for sub-CP power: Sweet Spot (~1–2h), Tempo (~2–4h), Endurance (~3–5h), etc., instead of a generic "sustainable for several hours".

► **Open Source License (GPL v3.0)**

The project is officially licensed under the GNU General Public License v3.0. Source code available at github.com/juancar1lo/ciclometricas.

3. What's New in Version 3.0

Version 3.0 adds new data editing tools, multi-language support, cross-training manual sessions, and stability improvements:

► **Power Spike Editor (new)**

New comprehensive module for detecting and correcting aberrant power spikes in .FIT and .TCX files. Three-phase pipeline: automatic detection (Z-score, absolute threshold, delta W/s), correction with four methods (linear interpolation, local average, cap, smart Auto), and export of the cleaned file. The Auto method automatically selects between interpolation and local average based on surrounding variability (coefficient of variation) and applies a physiological safety cap. Interface with interactive chart, individual spike cards with accept/reject, multi-channel context markers (power, HR, cadence), and detailed tooltips.

► **Manual Sessions (strength, walking, other)**

Log non-cycling activities from the import view or activity list. Each manual session includes activity type, duration, perceived IF (with presets by type), and automatic TSS calculation. Manual sessions are integrated into CTL/ATL/TSB calculations to reflect the total training load, not just cycling.

► **Torque (AET — Average Effective Torque)**

New module for calculating average effective torque ($AET = Power / (Cadence \times 0.10472)$). Available in the activity detail view to analyze the force-velocity relationship of pedaling.

► **Internationalization**

Complete internationalization system with support for Spanish (default) and English. Prepared for adding more languages in the future.

► **Health CSV Import/Export**

The Health module now includes CSV data import and export, compatible with Ciclométricas Web and any spreadsheet application (Excel, Google Sheets, LibreOffice). Bidirectional format with semicolon separator, automatic separator and date format detection, flexible header mapping (Spanish and English), and smart date-based upsert that prevents duplicates when importing multiple times.

4. Key Features

► **2-Parameter CP Model**

Linear regression (Monod-Scherrer) with CP, W', mFTP, VO₂max, sprint power, and Time to Exhaustion (TTE) estimation with detailed zone-based estimates. R² reliability indicator.

► **Power-Duration Curve (PDC)**

Global MMP with Savitzky-Golay smoothing, logarithmic scale 5s → 45min, interval range selection with targeting bands and suggested interval cards.

► **Durability Index (DRI)**

Empirical test of fresh vs. fatigued CP, exponential decay model with extrapolation, area-between-curves chart, and classification (excellent/good/improvable/limiting).

► **Load & Form (CTL/ATL/TSB)**

Exponential chronic and acute load model with forecast, weekly ramp rate calculation (Δ CTL), and Form chart with risk/optimal bands.

► **Power Zones (Coggan) & HR Zones (Friel)**

Coggan zones (8 power zones including Sweet Spot) and Friel zones (7 HR zones based on LTHR). Distribution by time, percentage, and range chart. Configurable reference: FTP, CP, or mFTP.

► **Fatigue Resistance**

FR index per activity with classification, temporal trend scatter chart, and detailed table.

► **Monotony (Foster)**

Weekly load and monotony with classification. Overload alerts.

► **Race Readiness (RRS)**

Score 0–100 based on form, load, consistency, and trend. Semicircular gauge with personalized advice.

► **Estimated Recovery**

Current state gauge, 7-day TSB projection, and recovery table per activity.

► **Health Module**

Tracking of resting HR, HRV, blood pressure, readiness, weight, body fat, and subcutaneous fat. Three interactive charts with tooltips, editable historical table, and multi-source readiness normalization.

► **W' Balance**

Skiba differential model (2012) — real-time visualization of W' expenditure and recovery during an activity.

► **Quadrant Analysis**

Coggan's analysis: pedaling distribution across 4 quadrants (neuromuscular, strength-endurance, CV efficiency, recovery).

► **Pedal Balance**

Left/right balance analysis with visual bar, symmetry classification, and dominant leg detection. Requires dual power meter.

► **Efficiency Trends**

Evolution of EF (NP/HR), VF (NP/Pavg), and Pw:Hr (power/HR decoupling) with three independent Y axes for precise analysis.

► **Interval Detection**

Automatic identification of work intervals with power, HR, cadence, and recovery time metrics.

► **Climb Detection**

Automatic identification of climbs with elevation gain, average gradient, power, and VAM.

► **Interactive Time Series**

Charts for power, HR, cadence, speed, altitude, and W' Balance with detailed tooltips.

► **Strava Integration**

OAuth 2.0 linking, automatic import of activities with all data streams.

► **Monthly PDF Report**

Generation of monthly reports with key metrics, zones, evolution, and summary.

► **Chart Export**

Any generated chart can be exported to various image formats, vector graphics, or CSV files.

► **Power Spike Editor**

Detection and correction of aberrant spikes in .FIT/.TCX files with smart Auto method, complete pipeline, and export of cleaned files.

► **Manual Sessions**

Logging of non-cycling activities (strength, walking, other) with perceived IF and TSS calculation integrated into the fitness model.

► **Torque (AET)**

Average Effective Torque calculation for force-velocity relationship analysis.

► **Internationalization**

Multi-language support (Spanish and English) with extensible i18n system.

► **Health CSV Import/Export**

Bidirectional health data exchange with Ciclométricas Web and spreadsheets, with smart date-based upsert.

5. Dashboard — Tabs

The Dashboard is the main view of Ciclométricas. It organizes all information into 6 thematic tabs, each with interactive charts, metric cards, and detailed tooltips.

5.1 Overview

Quick summary with key metrics for the selected period: total TSS, training hours, distance, total elevation gain, average power, average IF, and average Normalized Power. Cards with trend sparklines for at-a-glance evolution tracking.

5.2 Load & Form

Four main charts that monitor the cyclist's fitness state:

- **Daily Load:** daily TSS with 7-day moving average.
- **Fitness:** evolution of CTL (chronic fitness), ATL (acute fatigue), and TSB (balance).
- **Ramp Rate:** CTL increment per week (Δ CTL pts/week).

- **Form:** Form chart with bands: high risk (<-30), optimal (-30 to -10), productive (-10 to 5), fresh (5 to 20), and transition (>20).

5.3 Power and Heart Rate Zones

Time distribution across zones according to reference models:

- **Power:** Coggan zones (Z1 Recovery → Z7 Neuromuscular + Sweet Spot) with configurable reference (FTP, CP, or mFTP).
- **Heart Rate:** Friel zones (Z1 → Z5c) based on LTHR (Lactate Threshold Heart Rate). In v2.0, zones are calculated as a percentage of LTHR instead of maximum HR, following Joe Friel's updated methodology.
- **Visualization:** bar charts (absolute time and percentage), range table in watts/bpm, and horizontal range chart.

5.4 Performance

Critical power model analysis and derived metrics:

- **CP Model:** Critical Power (CP) and W' with R² reliability indicator.
- **Derived Metrics:** mFTP ($0.96 \times \text{CP}$), estimated VO₂max, sprint power.
- **TTE:** TTE calculator with detailed sub-CP zone-based estimates (just below CP ~1–2h, Sweet Spot ~2–4h, Tempo ~3–5h, Endurance >5h, Recovery many hours).
- **Trend:** historical evolution of CP and W' with trend chart.

5.5 Advanced Analysis

Three advanced sections to fine-tune training:

- **Efficiency Trends:** Evolution of EF (NP/HR), VF (NP/Pavg), and Pw:Hr (power/HR decoupling) with three independent Y axes: emerald for EF, amber for VF, and cyan for Pw:Hr. Each metric has its own scale for greater clarity.
- **PDC + Range Selection:** PDC curve in red with pink/violet targeting bands for planning intervals. Suggested range cards by duration (5s → 45min).

5.6 Workouts

Visual monthly calendar with month-by-month navigation. Each day shows TSS, duration, and color based on intensity. Allows accessing any activity's detail with a single click.

6. Specialized Modules

In addition to the Dashboard, Ciclométricas includes independent modules accessible from the sidebar, each dedicated to a specific aspect of performance.

6.1 Training Summary

Weekly and monthly aggregate tables with sparklines for TSS, hours, distance, elevation, average IF, and average NP. Quick visualization of medium- and long-term trends. Monthly PDF report generation.

6.2 Monotony (Foster, 1998)

Implementation of Foster's model to detect potentially dangerous load patterns:

- **Calculation:** Monotony = $\text{mean_TSS} / \text{standard_deviation_TSS}$ per week.
- **Strain:** Strain = $\text{weekly_TSS} \times \text{Monotony}$.
- **Classification:** Normal (<1.5), Elevated (1.5–2.0), Dangerous (>2.0).
- **Visualization:** Weekly load bar chart + monotony line + detailed table with color-coded classification.

6.3 Race Readiness (RRS)

Score from 0 to 100 that synthesizes the cyclist's readiness to race. Combines four weighted factors:

- **Form:** current form (TSB in optimal zone).
- **Fitness:** accumulated chronic load (CTL).
- **Consistency:** recent training regularity.
- **Trend:** trend over the last 14 days.

Includes semicircular gauge, factor bars, and personalized text advice.

6.4 Fatigue Resistance

FR (Fatigue Resistance) index per activity: measures the ability to maintain power throughout a long activity. Classification from Excellent to Limiting. Temporal evolution scatter chart with trend line.

6.5 Durability (DRI)

Comprehensive module for evaluating durability — the ability to maintain performance after accumulating work (kilojoules):

- **Test:** recording empirical tests: 3min and 12min power after X kJ of fatigue.
- **Calculation:** $\text{DRI} = (\text{fatigued_CP} / \text{fresh_CP}) \times 100$. Classification: excellent (>90%), good (80–90%), improvable (70–80%), limiting (<70%).
- **CP Curves:** chart of fresh vs. fatigued CP curves.
- **Extrapolation:** $\text{DRI}(\text{kJ}) = 100 \times e^{(-\lambda \times \text{kJ})}$ model to predict durability at any workload volume.
- **Area Between Curves:** area chart between fresh and fatigued curves with power differential.

6.6 Recovery

Circular gauge with current recovery state, CTL/ATL/TSB metrics, 7-day TSB projection chart, and estimated recovery table per activity.

6.7 Health (new in v2.0)

New comprehensive health tracking module, accessible from the sidebar. Track and visualize the following metrics over time:

- **Cardiovascular:** Resting HR (bpm)
- HRV (ms) — heart rate variability
- Readiness — normalized to 1-10 scale from multiple sources
- **Blood Pressure:** Systolic and diastolic blood pressure (mmHg)
- **Body Composition:** Weight (kg), body fat (%), subcutaneous fat (mm)

Three interactive charts with tooltips:

- **Cardiovascular:** Resting HR + HRV (dual axes) with readiness line and color-coded reference bands (Low, Moderate, Good, Optimal).
- **Blood Pressure:** candlestick-style chart with diastolic → systolic candles and color-coded classification (normal, elevated, high, very high).
- **Body Composition:** weight, body fat, and subcutaneous fat with independent axes.

Supported readiness sources: manual (1-10), Garmin (1-100), Whoop (1-100), Oura (1-100), COROS (1-100), Elite HRV (1-10). All are automatically normalized to the unified 1-10 scale.

Historical table with editing and deletion of any record.

Health data import and export in CSV format (new in v3.0), compatible with Ciclométricas Web and spreadsheets. Smart date-based upsert when importing.

6.8 Spike Editor (new in v3.0)

Module for detecting and correcting aberrant power spikes, for occasional use when erroneous data is suspected. Workflow: open file → automatic detection → spike-by-spike review → accept/reject → export cleaned file.

► Detection:

three configurable criteria: Z-score (adjustable sensitivity), absolute maximum watts threshold, and power delta per second (W/s).

► Correction:

four methods: linear interpolation, local average, cap (limit), and Auto (smart selection based on the surrounding coefficient of variation with a physiological safety cap).

► Interface:

interactive chart with green/red markers, individual spike cards with detection detail and original vs. corrected values, multi-channel tooltips (power, HR, cadence), and impact table.

► Export:

generates a new clean .FIT or .TCX file with the accepted corrections, preserving the rest of the original file data.

7. Activity Detail

Each imported activity has a comprehensive detail view with:

- **Key Metrics:** duration, distance, elevation, TSS, IF, NP, average power, average/max HR, average cadence.
- **Time Series:** charts for power, HR, cadence, speed, altitude, and W' Balance with individual toggles.
- **Quadrant Analysis:** pedaling distribution across Coggan's 4 quadrants. Crank length selection (from 165 to 180mm).
- **Pedal Balance (new in v2.0):** Visual bar with symmetry classification (balanced, slight asymmetry, notable asymmetry) and dominant leg detection. Requires dual-sensor power meter.
- **Work Intervals:** automatic detection with power, HR, cadence, and recovery per interval.
- **Climbs:** automatic detection with elevation gain, gradient, power, and VAM.
- **MMP (Power Records):** best efforts at standard durations (5s, 1min, 5min, 20min, 60min...).

- **Route Map:** map with GPS route colored by power/HR.
- **Notes:** editable notes field and customizable name.

► **Manual Sessions:**

non-cycling activities show perceived IF, TSS formula, and activity type. Cycling-specific sections are automatically hidden.

► **Torque (AET):**

average effective torque calculated as $\text{Power} / (\text{Cadence} \times 0.10472)$, expressed in Nm.

8. Data Import

Ciclométricas supports two import methods:

File Import

- .FIT files (Garmin, Wahoo, Hammerhead, COROS, etc.)
- .TCX files (Garmin Connect, Strava export, etc.)
- Drag & drop or multiple file selection.
- Automatic processing: calculation of metrics, zones, MMP, intervals, climbs, and pedal balance.
- Duplicate detection by date and duration.
- Left/right balance parsing from native FIT SDK format.

Import from Strava

- Secure OAuth 2.0 linking with your Strava account.
- Automatic download of complete streams (power, HR, cadence, altitude, GPS, speed).
- Smart filtering of cycling activities (Road, MTB, Gravel, Virtual, E-Bike...).
- Automatic pagination (up to 250 activities).
- Strava ↔ Strava and Strava ↔ .FIT/.TCX duplicate detection.
- Stable threading architecture (Python threading + QTimer polling) that prevents freezing.

9. Strava Integration

Ciclométricas integrates directly with Strava through its official v3 API. The linking flow is:

1. Create an application at strava.com/settings/api (free).
2. Enter the Client ID and Client Secret in Ciclométricas.
3. Authorize access in the browser (redirect to localhost:5765).
4. Cycling activities are imported with a single click, with all original data.

Tokens are automatically renewed. You can unlink and relink at any time without losing previously imported activities.

10. Settings & Rider Profile

The settings tab allows customization of:

- **Power Threshold:** FTP (Functional Threshold Power).
 - **Max HR:** maximum heart rate.
 - **LTHR (new in v2.0):** Lactate Threshold Heart Rate (LTHR). In v2.0, HR zones are calculated as a percentage of LTHR.
 - **Weight:** athlete weight (for VO_2max and W/kg).
 - **Zone Source:** FTP, CP, or mFTP as the zone basis.
 - **Power Tests:** recording power tests (short, medium, long) to calibrate the CP model.
 - **History:** history of saved configurations.
- **Language:**
interface language selection (Spanish or English). New in v3.0.

11. Requirements & Installation

System Requirements:

- Python 3.11 or higher
- Windows 10/11, Linux (Ubuntu 22.04+, Fedora 38+, etc.)
- ~800 MB of disk space
- Internet connection only for Strava integration

Installation:

- 1. Download the .zip file from GitHub or GitHub Releases
- 2. Extract to a folder
- 3. Install dependencies: `pip install -r requirements.txt`
- 4. Run: `python main.py`

It can also be installed with pyproject.toml: `pip install .` and run as: `ciclometricas`

For regular users, through an executable .exe installer.

12. Technology & Architecture

Component	Technology
GUI	PySide6 (Qt 6)
Interactive Charts	PyQtGraph + NumPy + SciPy
Local Database	SQLite + SQLAlchemy ORM
File Parsing	fitdecode (FIT) + lxml (TCX)
Strava Integration	OAuth 2.0 + API v3 REST
Reports	ReportLab (PDF)
Tests	pytest (223 tests)

Ciclométricas is an independent project, developed with rigor and passion for cycling and the science of sports performance. All data is processed locally — your data never leaves your computer.

13. License

Ciclométricas is distributed under the GNU General Public License v3.0 (GPL-3.0). This means you can freely use, modify, and distribute the software, provided that any derivative work is distributed under the same license and the source code remains accessible.

Source code: github.com/juancar1lo/ciclometricas

 **Ciclométricas — Your cycling performance lab, on your desktop.**

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